



Вселенная

[Главная](#)

[Книги](#)

[О нас](#)

[Контакты](#)

[Политика конфиденциальности](#)

[DMCA](#)

Google Cloud Platform для анализа данных

Авторы: доктор Шиталкумар Р. Сухдев, Сандика С. Сухдев

Тип файла: pdf

Размер: 4,6 МБ

Язык: Английский

Страниц: 219

Google Cloud Platform для Data Science: краткий курс по большим данным, машинному обучению и сервисам анализа данных

Введение

Наука о данных прошла путь от анализа электронных таблиц на отдельных компьютерах до обработки огромных массивов данных, распределенных по глобальной облачной инфраструктуре. Для студентов, исследователей, инженеров и профессиональных специалистов по анализу данных **Google Cloud Platform (GCP)**

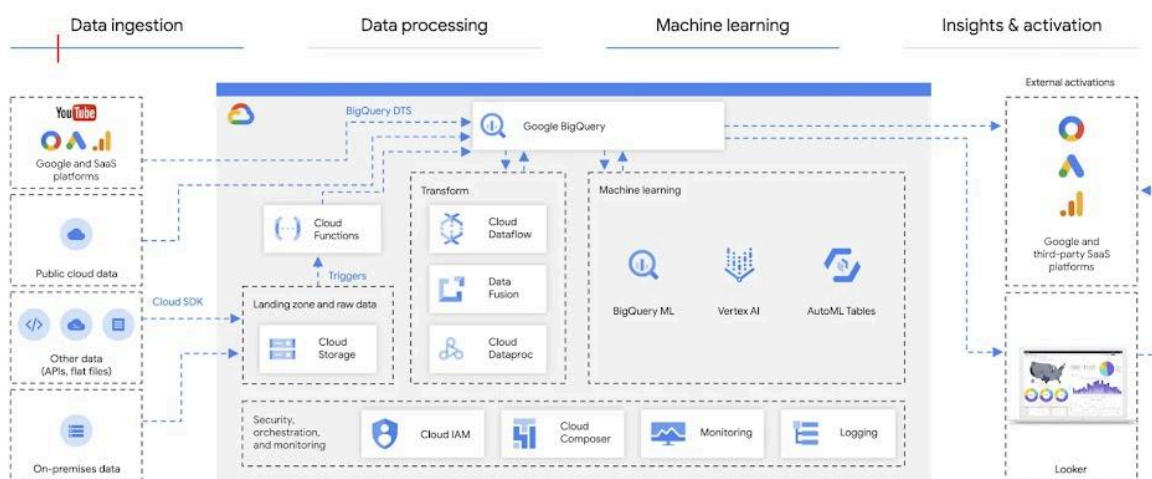
предоставляет комплексную среду для сбора, хранения, обработки, анализа и моделирования данных в любом масштабе.

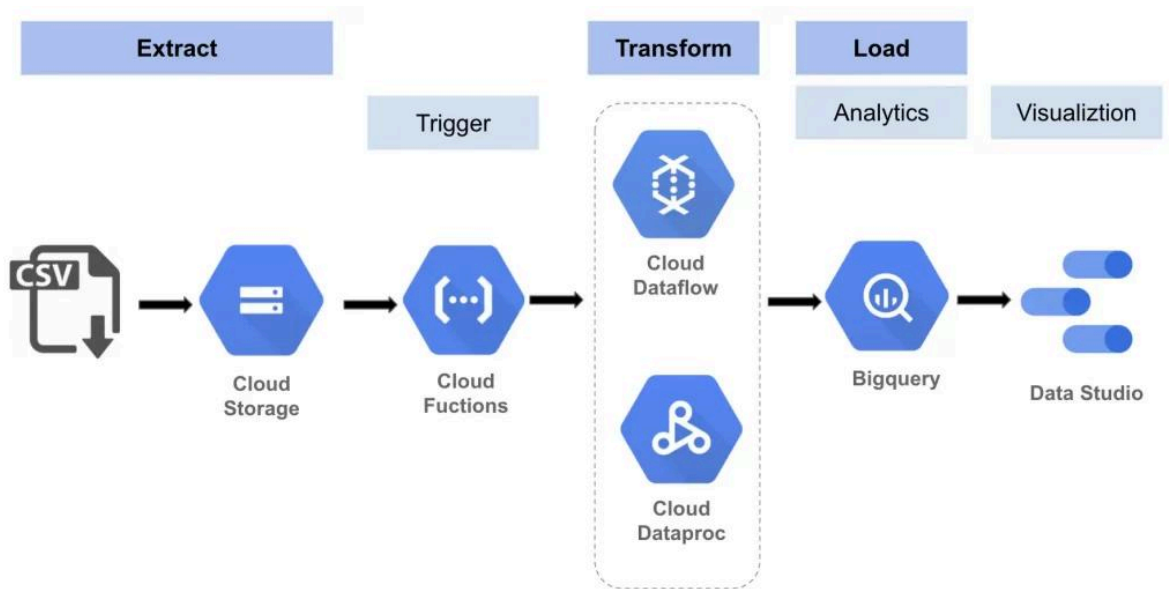
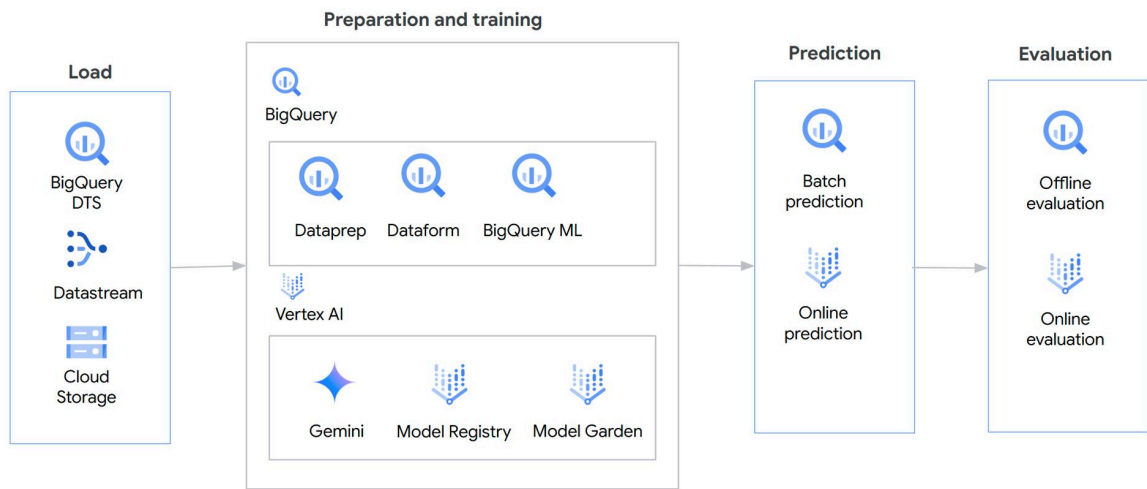
Типичный проект в области науки о данных может начинаться с необработанных данных, полученных от приложений, датчиков, веб-сайтов, финансовых систем или промышленного оборудования. Затем эти данные проходят несколько этапов обработки:

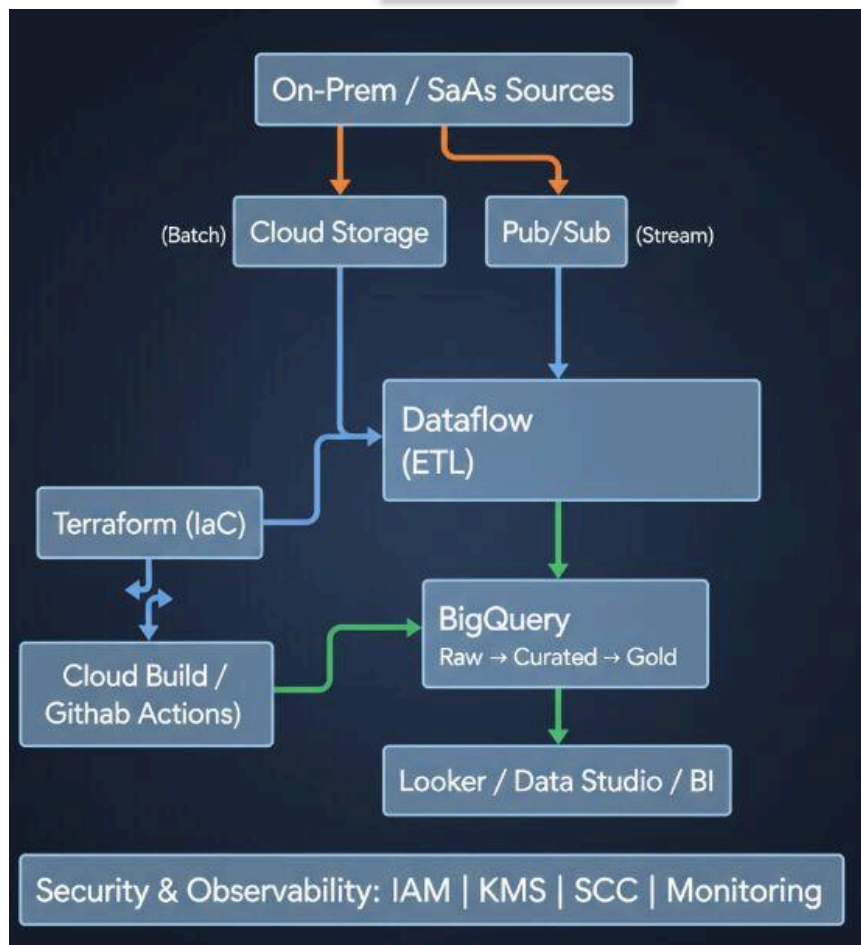
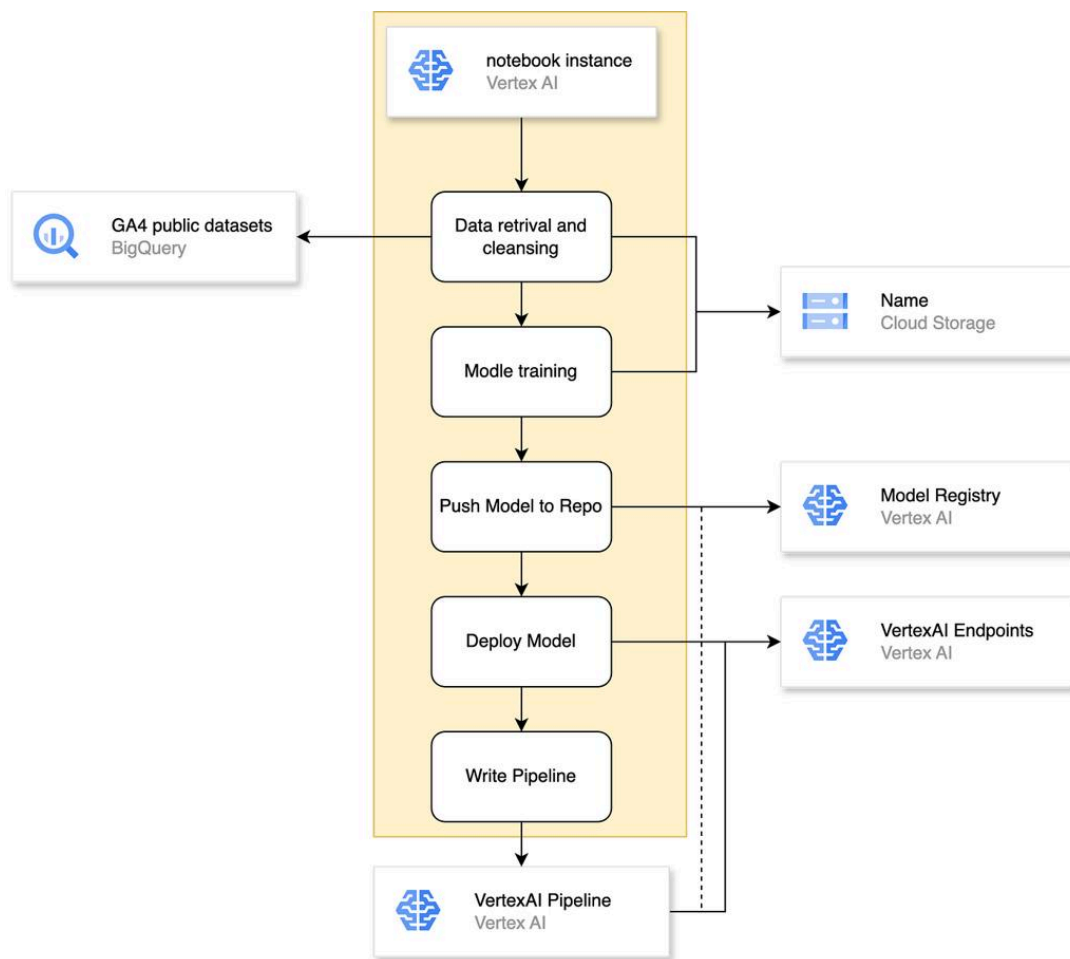
Источники данных → Хранение → Обработка → Аналитика → Машинное обучение → Развертывание → Мониторинг

Google Cloud предлагает сервисы практически для каждого этапа этого конвейера. **BigQuery** предоставляет масштабируемый аналитический SQL, **Cloud Storage** — объектное хранилище, **Dataflow** — поддержку пакетной и потоковой обработки, **Dataproc** — управляемые среды Spark и Hadoop, а **Vertex AI** — инструменты для разработки и развертывания систем машинного обучения.

Цель этого экспресс-курса — дать практическое представление об этих технологиях, не предполагая, что читатель уже является экспертом в области облачных вычислений. 🚀



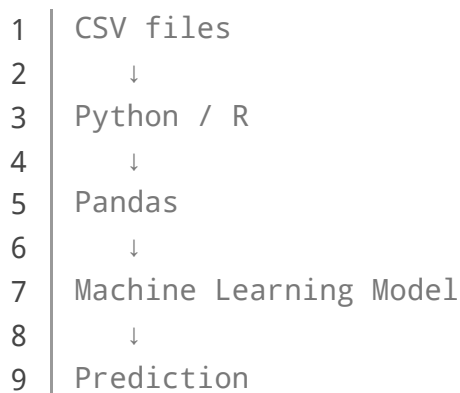




Теоретическая основа

Почему облачные вычисления важны для науки о данных

Традиционная наука о данных часто начинается с локальной рабочей станции:



Этот подход хорошо работает с небольшими наборами данных. Однако проблемы возникают, когда объем данных становится слишком большим для локальной памяти или когда доступ к одной и той же инфраструктуре нужен нескольким пользователям.

Предположим, инженерная организация генерирует **10 ТБ данных с датчиков каждый месяц**.

Обычный компьютер может не справиться с:

- Объем хранилища
- Ограничения оперативной памяти
- Время обработки
- Требования к резервному копированию
- Параллельные вычисления
- Совместная работа
- Развертывание модели

Облачные вычисления устраняют эти ограничения за счет распределения рабочих нагрузок по масштабируемой инфраструктуре.

Распределенная обработка данных

В основе современной облачной аналитики лежит **распределенная обработка данных**.

Вместо обработки:

$$D=D1+D2+D3+\dots+Dn$$

на одном компьютере, распределенная платформа делит набор данных на разделы и обрабатывает их одновременно.

Если:

T_{single} = время обработки на одном компьютере

и N рабочих процессов обрабатывают набор данных параллельно, то теоретическое время обработки приближается к:

$T_{\text{parallel}} \approx N T_{\text{single}}$

Реальные системы не являются идеально масштабируемыми из-за накладных расходов на коммуникацию, синхронизацию, хранение данных и планирование.

Теория конвейера обработки данных

Производственная система обработки данных обычно состоит из нескольких уровней:

1. **Data ingestion**
2. **Data storage**
3. 🌈 **Data transformation**
4. **Data analytics**
5. **Feature engineering**
6. **Machine learning**
7. **Model deployment**
8. **Monitoring**

GCP provides specialized services for each layer.

Definition

What is Google Cloud Platform for data science?

Google Cloud Platform is a collection of cloud computing services that can be used to build scalable systems for **data engineering, data analytics, artificial intelligence, and machine learning**.

For data scientists, the most important services include:

GCP Service	Primary Purpose
Cloud Storage	Store files and datasets
BigQuery	Large-scale analytical database
Dataflow	Batch and streaming processing
Dataproc	Managed Spark/Hadoop
Pub/Sub	Real-time messaging
Vertex AI	Machine learning platform
Cloud SQL	Relational databases
Looker	Business intelligence and visualization
Data Catalog / governance tools	Data discovery and management
Cloud Functions / Cloud Run	Application and processing workloads

BigQuery

BigQuery is one of the most important GCP services for data scientists.

It is designed for analytical workloads using SQL.

For example:

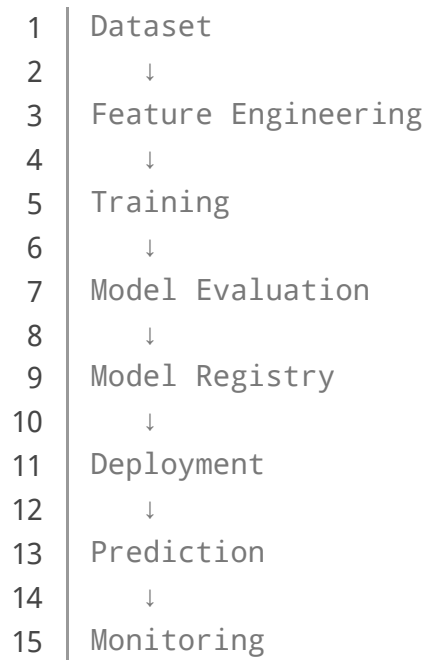
```
1 | SELECT
2 |     country,
3 |     AVG(revenue) AS average_revenue
4 | FROM `project.analytics.sales`
5 | GROUP BY country
6 | ORDER BY average_revenue DESC;
```

A data scientist can perform large-scale analysis without manually managing database servers.

Vertex AI

Vertex AI provides a managed environment for machine-learning workflows.

A simplified architecture is:



It can support traditional machine learning as well as modern AI workflows.

Step-by-Step Explanation

Step 1: Define the data problem

Before opening the Google Cloud console, define the engineering question.

For example:

Can we predict whether an industrial machine will fail within the next seven days?

The target variable might be:

$Y = \{1 \text{ machine failure}, 0 \text{ normal operation}\}$

Potential features include:

- Temperature
- Vibration
- Pressure
- Operating hours

- Motor speed
- Previous maintenance
- Electrical current

Step 2: Collect the data

Data can originate from:

- IoT sensors
- Applications
- Databases
- APIs
- CSV files
- Logs
- Enterprise systems

For files and raw datasets, **Cloud Storage** can act as the data lake.

A typical structure might be:

```
1 | gs://engineering-data/  
2 |   raw/  
3 |   cleaned/  
4 |   features/  
5 |   models/  
6 |   reports/
```

Step 3: Load data into BigQuery

After the raw data is stored, structured datasets can be loaded into BigQuery.

The analytical workflow becomes:

```
1 | Cloud Storage  
2 |   ↓  
3 |   BigQuery  
4 |   ↓  
5 |   SQL Analysis  
6 |   ↓  
7 |   Feature Dataset
```

Step 4: Explore the dataset

A data scientist might calculate:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

for the mean sensor value.

Variance can be calculated as:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

These basic statistics help identify abnormal measurements.

Step 5: Build features

Raw measurements are often not suitable directly for machine learning.

Suppose vibration is recorded every minute.

Instead of using only:

$$V_t$$

we can create:

$$V_{avg,24h} \quad V_{max,24h} \quad \Delta V = V_t - V_{t-1}$$

These features can provide more useful information about equipment behavior.

Step 6: Train the machine-learning model

A classification model might estimate:

$$P(Y=1|X)$$

where X represents the machine's observed characteristics.

Possible models include:

- Logistic regression
- Decision trees
- Random forests
- Gradient boosting
- Neural networks

Step 7: Evaluate the model

Accuracy alone is often insufficient.

Important metrics include:

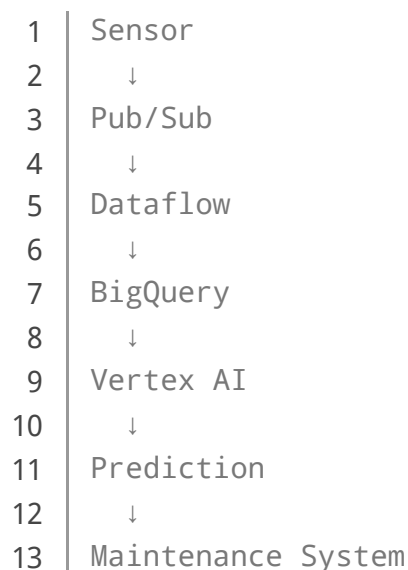
$\text{Precision} = \frac{TP}{TP + FP}$ $\text{Recall} = \frac{TP}{TP + FN}$ $F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

For predictive maintenance, recall may be especially important because missing an actual machine failure can be expensive.

Step 8: Deploy the model

After validation, the model can be deployed through Vertex AI.

The production architecture could look like:



Step 9: Monitor performance

Machine-learning systems change over time.

A model trained on last year's operating conditions may perform poorly after equipment, production processes, or customer behavior changes.

This phenomenon is commonly associated with **data drift** and **concept drift**.

Comparison

GCP services for data science

Service	Best For	Typical User
BigQuery	Analytics and SQL	Data Scientist
Cloud Storage	Data lake/file storage	Data Engineer
Dataflow	Streaming/batch pipelines	Data Engineer
Dataproc	Spark/Hadoop workloads	Data Engineer
Vertex AI	ML development/deployment	ML Engineer
Pub/Sub	Real-time events	Cloud Engineer
Looker	Business analytics	Analyst
Cloud SQL	Transactional relational data	Developer

BigQuery vs traditional databases

Traditional relational databases are often optimized for transactional workloads.

For example:

```
1 | INSERT
2 | UPDATE
3 | DELETE
```

Analytical platforms are optimized for queries involving enormous numbers of rows:

```
1 | GROUP BY
2 | JOIN
3 | SUM
4 | AVG
5 | COUNT
6 | WINDOW FUNCTIONS
```

The distinction between **OLTP** and **OLAP** is therefore important.

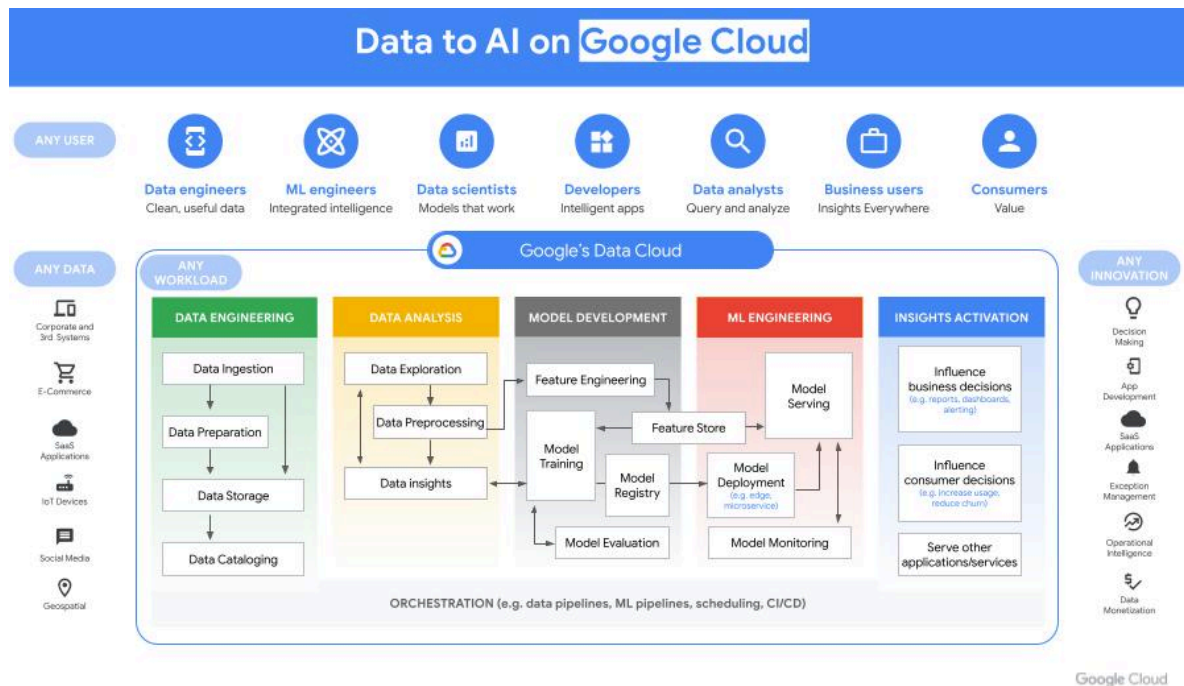
Dataflow vs Dataproc

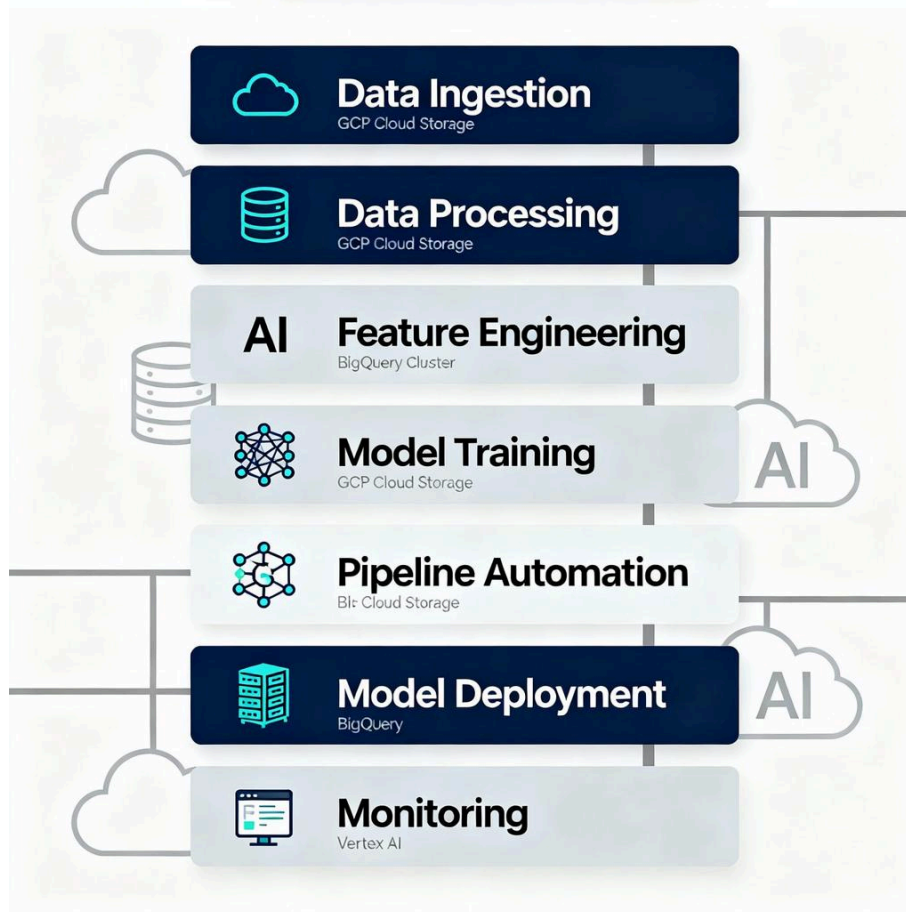
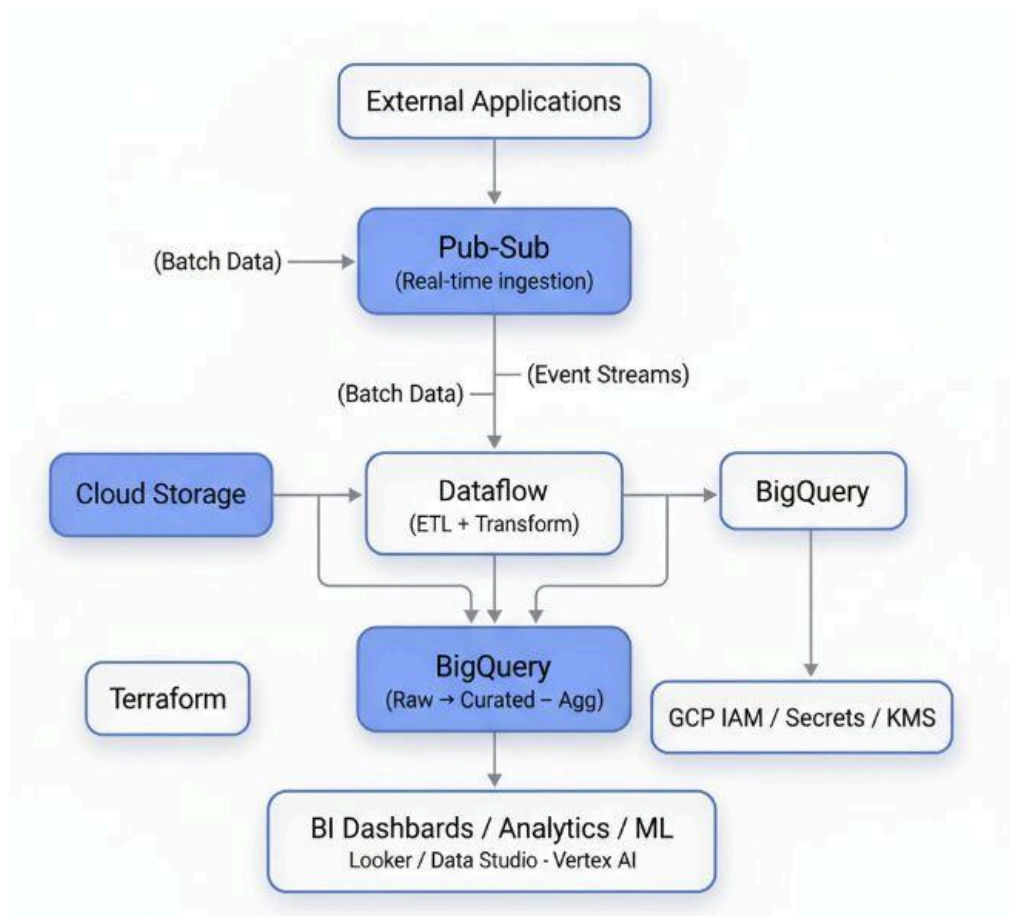
Dataflow is particularly useful when you want managed Apache Beam pipelines for batch and streaming processing.

Dataproc is useful when your workload specifically benefits from Apache Spark, Hadoop, or related ecosystem technologies.

Diagrams & Tables

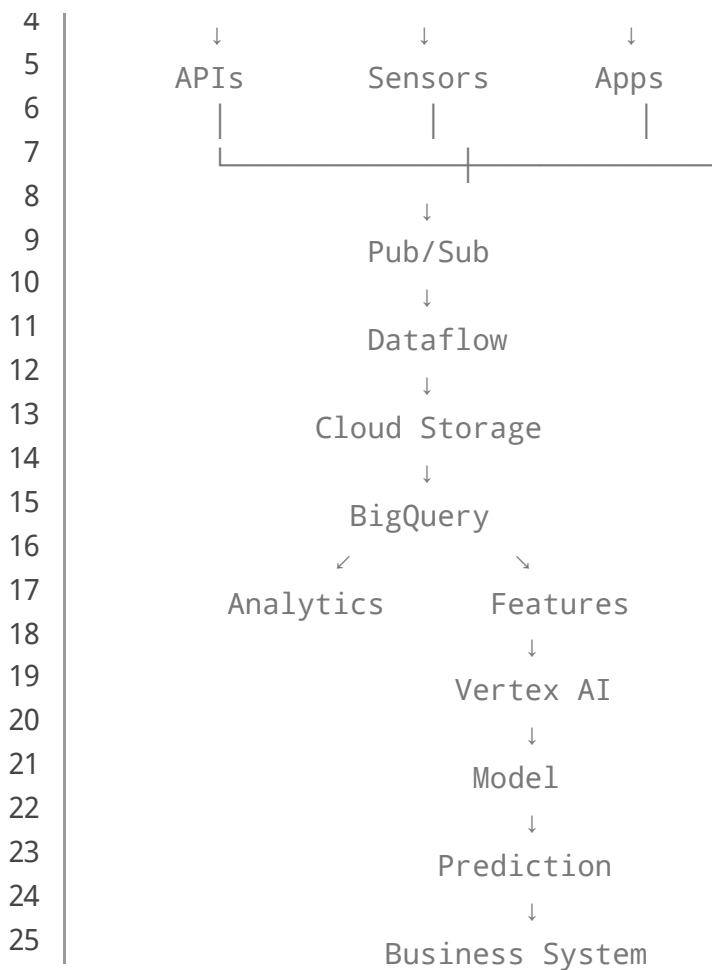
Complete GCP data-science architecture





A simplified architecture is:





Data lifecycle

Stage	Question	GCP Technology
Collection	Where does data originate?	Pub/Sub, APIs
Storage	Where is raw data kept?	Cloud Storage
Processing	How should it be transformed?	Dataflow/Dataproc
Analytics	What happened?	BigQuery
Modeling	What will happen?	Vertex AI
Visualization	How do users understand results?	Looker
Deployment	How does the model provide predictions?	Vertex AI
Monitoring	Is the system still reliable?	Cloud monitoring/ML monitoring

Examples

Example 1: Customer churn prediction

An organization wants to predict whether a customer will cancel a service.

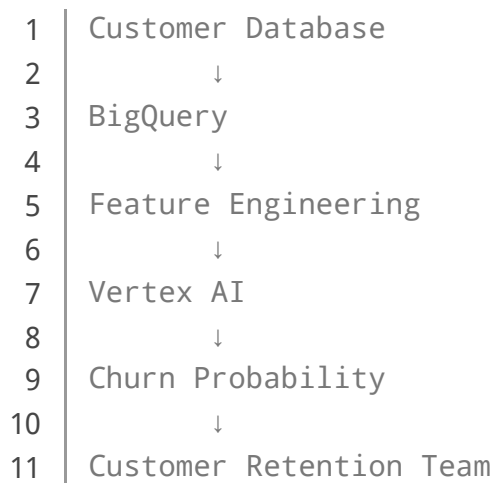
Input variables:

$X=[\text{tenure, usage, complaints, payments, plan}]$

Output:

$Y \in \{0, 1\}$

The workflow becomes:



Example 2: Sales forecasting

Historical sales can be analyzed by:

- Product
- Region
- Date
- Customer segment
- Marketing campaign

The objective might be:

$\text{Sales}_{t+1} = f(\text{Sales}_t, \text{Seasonality}, \text{Price}, \text{Marketing})$

The model can then generate future demand estimates.

Example 3: Engineering anomaly detection

Suppose a factory generates:

500 GB of sensor data every day.

Instead of manually reviewing every measurement, an anomaly-detection system can identify unusual patterns.

For a measurement x , a standardized score can be represented as:

$$z = \frac{x - \mu}{\sigma}$$

Very large positive or negative values can indicate unusual behavior, although production systems generally require more sophisticated approaches.

Real-World Application

Smart manufacturing

Manufacturing organizations can combine IoT data, cloud analytics, and machine learning to develop predictive-maintenance systems.

Sensors continuously generate:

- Temperature
- Pressure
- Vibration
- Voltage
- Current
- RPM

Streaming infrastructure can process these events.

Analytics platforms can store historical information, while machine-learning models can estimate failure probabilities.

Financial analytics

Financial institutions can use cloud data platforms for:

- Fraud detection
- Risk analysis
- Customer segmentation

- Forecasting
- Transaction monitoring

However, financial workloads require strict security, governance, access control, and regulatory compliance.

Healthcare analytics

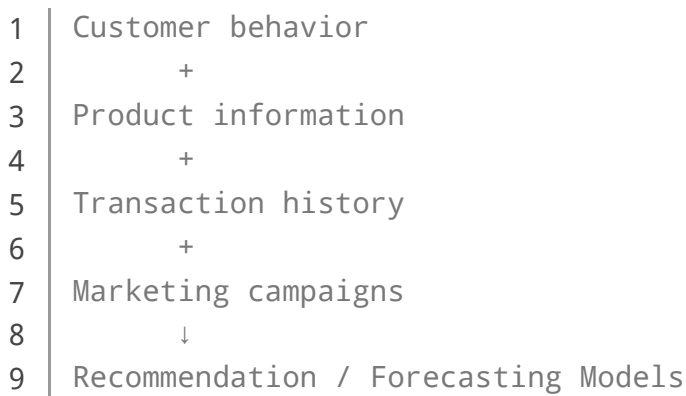
Cloud analytics can support:

- Medical research
- Population analysis
- Operational forecasting
- Image analysis
- Clinical research

Sensitive healthcare data requires particularly careful handling of privacy, security, and applicable regulations.

Retail

Retail organizations can analyze:



This can support inventory planning and personalized recommendations.

Common Mistakes

Using BigQuery as a replacement for every database

BigQuery is excellent for analytical workloads, but that does not mean every application should use it as its primary transactional database.

Choose the storage system according to workload requirements.

Ignoring data quality

A sophisticated model trained on poor-quality data will still produce poor results.

Remember:

Garbage In→Garbage Out

Training before understanding the data

Jumping immediately into machine learning can create unnecessary complexity.

First examine:

- Missing values
- Duplicates
- Outliers
- Data distributions
- Correlations
- Label quality

Using accuracy blindly

If only 1% of transactions are fraudulent, a model that predicts “not fraud” every time can achieve 99% accuracy while being practically useless.

Forgetting cloud costs

Cloud infrastructure is scalable, but scalability does not mean unlimited free computation.

Large queries, continuous pipelines, storage, and model-serving infrastructure can generate substantial costs.

Challenges & Solutions

Challenge: Large datasets

Problem: Processing billions of records locally is impractical.

Solution: Use distributed processing and analytical services such as BigQuery, Dataflow, or Dataproc.

Challenge: Real-time data

Problem: Batch processing introduces delays.

Solution: Use event-driven architectures involving Pub/Sub and streaming processing.

Challenge: Model degradation

Problem: Production data changes.

Solution: Monitor model performance and data distributions, and establish retraining policies.

Challenge: Security

Problem: Data science environments may contain confidential information.

Solution: Apply least-privilege IAM, encryption, appropriate network controls, auditing, and data-governance practices.

Challenge: Complex architectures

Problem: Beginners may attempt to use every available cloud service.

Solution: Start with the simplest architecture that satisfies the requirements.

Case Study

Predictive maintenance for a manufacturing plant

Consider a hypothetical manufacturing company operating **2,000 industrial machines**.

Each machine produces sensor data every minute.

The organization wants to predict failures before they occur.

Data collection

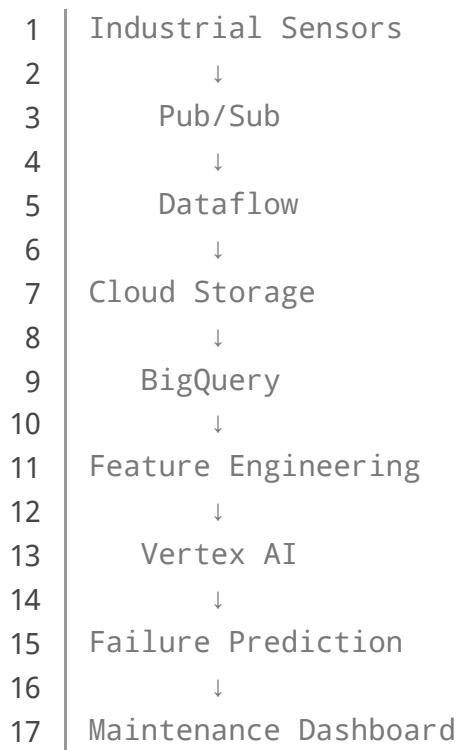
Approximately:

$$2,000 \times 1,440 = 2,880,000$$

sensor records could be generated each day for just one measurement per minute per machine.

If every machine generates multiple measurements, the dataset becomes significantly larger.

Architecture



Business result

Suppose the organization historically experiences:

100 failures/month

If predictive maintenance allows engineers to identify a meaningful portion of failures early, maintenance can be scheduled before catastrophic breakdowns occur.

The real benefit is not simply a high model accuracy score.

It is the reduction of:

- Downtime
- Emergency maintenance
- Production losses
- Equipment damage
- Unplanned labor costs

This illustrates an important engineering principle:

The value of a machine-learning model is determined by the decision it improves, not merely by its statistical sophistication.

Essential Tips

Start with BigQuery and Cloud Storage

For beginners, these two services provide a strong foundation.

Learn how to:

1. Upload data.
2. Create datasets.
3. Write SQL.
4. Build analytical queries.
5. Optimize queries.
6. Export results.

Learn SQL seriously

Data scientists working with cloud platforms should not depend entirely on Python.

Strong SQL skills are essential for:

- Filtering
- Joining
- Aggregation
- Window functions
- Feature engineering
- Data validation

Learn Python alongside cloud technologies

Python remains valuable for:

- 1 | Python
- 2 | | NumPy
- 3 | | Pandas
- 4 | | Scikit-learn
- 5 |

Think in pipelines

Instead of asking:

“How do I train a model?”

ask:

“How does data move from its source to a reliable prediction?”

That change in perspective is fundamental to production data science.

Optimize before scaling

Do not immediately deploy a large cluster.

First ask:

- Can SQL solve the problem?
- Can BigQuery handle the workload?
- Do I actually need Spark?
- Is streaming required?
- Does the model need real-time inference?

Simple architectures are generally easier to maintain.

FAQs

What is [Google Cloud Platform](#) used for in [data science](#)?

GCP can be used to store datasets, process large-scale data, perform analytics, train machine-learning models, deploy models, and monitor production systems.

Is BigQuery a machine-learning platform?

BigQuery is primarily a cloud data warehouse and analytical platform. It also provides machine-learning capabilities through BigQuery ML, allowing certain models to be created directly using SQL.

Is Vertex AI difficult for beginners?

Vertex AI contains advanced capabilities, but beginners can approach it progressively. Start with datasets, basic model training, evaluation, and deployment before moving into advanced MLOps.

Do I need Python to use GCP for data science?

Python is extremely useful but not always mandatory. BigQuery allows substantial analytical work using SQL, while Python becomes particularly valuable for advanced data processing and machine learning.

What is the difference between BigQuery and Cloud Storage?

Cloud Storage is object storage designed for files and unstructured or semi-structured data. BigQuery is designed primarily for high-performance analytical querying.

Should I learn Dataflow or Dataproc first?

For many beginners, learning BigQuery first is more practical. After understanding data warehouses and SQL, choose Dataflow when you need Apache Beam-based batch/streaming pipelines and Dataproc when Spark/Hadoop workloads are appropriate.

Can GCP handle real-time machine learning?

Yes. A GCP architecture can combine streaming ingestion, processing, feature generation, model serving, and monitoring to support real-time or near-real-time machine-learning applications.

Is cloud computing necessary for data science?

No. Many data-science concepts can be learned locally. However, cloud platforms become increasingly valuable when datasets, computational requirements, collaboration, deployment, or production workloads grow.

Conclusion

Google Cloud Platform provides a powerful ecosystem for modern data science, combining **large-scale analytics, distributed processing, machine learning, data engineering, and production deployment**.

The most important concept is not memorizing the names of dozens of cloud services. Instead, understand how they fit into a complete engineering workflow:

Data→Storage→Processing→Analytics→Features→ML→Deployment→Monitoring

For beginners, a sensible learning path is:

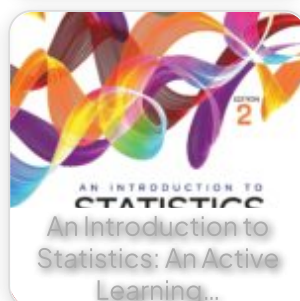
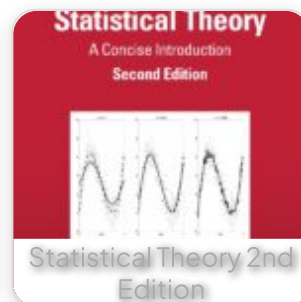
Cloud Storage → BigQuery → SQL → Dataflow/Dataproc → Vertex AI → MLOps

For experienced engineers, the challenge shifts from simply building models to designing systems that are **scalable, secure, observable, cost-efficient, and maintainable**.

Ultimately, GCP for data science is less about a single technology and more about building an integrated data-to-decision system. 🌐 📊 🤖

When these components are designed correctly, enormous datasets can move from raw sensor readings, transactions, logs, or customer interactions to actionable predictions—turning cloud infrastructure into a practical engineering platform for modern intelligent systems.

Related Posts:



Preview PDF Book

[← Previous Book](#)

[Next Book →](#)

EngiVerse

Explore a vast library of free engineering ebooks curated to help you expand your technical knowledge, conduct groundbreaking research, and advance your career. Our mission is to create a hub of knowledge that fosters innovation, empowers aspiring engineers, and bridges the gap between education and industry.

[Home](#) [Books](#) [About Us](#) [Contact](#)
[Privacy Policy](#) [DMCA](#)

copyright 2025 EngiVerse. All Right Reserved